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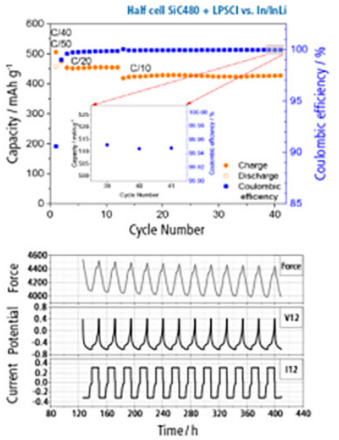
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High Sensitive Voltammetric Determination of Betamethasone on an Amalgam Film Electrode

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A voltammetric method for fast and high sensitive determination of betamethasone (BM) using an amalgam film silver based electrode (Hg(Ag)FE) was developed. Cyclic voltammetry study revealed that the BM reduction process is diffusion controlled with the participation of two electrons and two protons. In the optimum conditions, calibration curve was linear for BM in the concentration range from $5 \cdot 10^{-9}$ mol L^{-1} ($1.96 \cdot 10^{-6}$ g L^{-1}) to $0.8 \cdot 10^{-6}$ mol L^{-1} ($3.14 \cdot 10^{-4}$ g L^{-1}), with a detection limit of $1.6 \cdot 10^{-9}$ mol L^{-1} for preconcentration time of 45 s. Repeatability of method was determined as RSD % for betamethasone concentration of $0.1 \cdot 10^{-7}$ mol L^{-1} as 2.1% (n = 5). The proposed method was successfully applied and validated by studying the recovery of betamethasone in commercially available drugs.
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Contemporary medicine widely use drugs containing synthetic steroids during treatment of many conditions. Due to its popularity in different medicaments, quantitative analysis of its content is one of the most important issues in modern analysis. With still increasing number of drugs used in therapies, the development of new, high sensitive analytical methods is required. The most popular quality control techniques includes chromatographic and mass spectroscopic methods.[1–3] However, electroanalytical methods are also gaining much attention due to its simplicity and sensitivity.[4–6]

Betamethasone (BM) is the representative of the synthetic steroids, belonging to the glucocorticoid class, frequently used in the widespread clinical applications due to its anti- inflammatory and immunosuppressant activity. It is commonly used medication in prenatal medicine inter alia in order to accelerate advance lung maturation in fetus of pregnant woman at risk of premature delivery,[7] to decrease the mortality incidences from intracranial haemorrhage during premature infants treatment[8] and for reduction of neonatal respiratory distress.[9–11] Besides, it is widely used in treatment of dermatology conditions, such as psoriasis vulgaris,[12–14] vitiligo[15] or eczema.[16–18] It is also very effective drug used in the therapy of asthma,[19–21] allergy,[22,23] rheumatoid arthritis[24–26] and many others.[27–30] Considering wide range of betamethasone therapeutic usage, development of its sensitive determination methods is highly required.

For high sensitive betamethasone determination in drug samples and biological matrices, mainly chromatographic and mass spectrometric methods are used.[31–46] These methods provides high sensitivity of measurements but also require time consuming sample preparation, expensive instrumentation and reagents. To reduces these disadvantages but still maintain the sensitivity of measurements, voltammetric methods of betamethasone determination have been proposed. In the available literature there is a few reports of such measurements, including use of hanging mercury drop electrode[47] and single wall carbon nanotube modified edge plane pyrolytic graphite electrode[48,49] as the working electrode, where the modification by use of nanotubes caused improvement of electrode parameters.

To achieve low betamethasone detection limit, the renewable amalgam film electrode (Hg(Ag)FE)[50] has been used in this study. Among the variety of available film electrode constructions, we can distinguish the most popular ones, consisted of glassy carbon disc or silver rod as the electrode base, covered by bismuth, lead, gallium or mercury film.[51–53] For further measurements, film electrode consisted of silver rod covered by mercury was used. This particular type of film electrode is especially attractive, merging the advantages of classical mercury electrodes with high working surface, which is characteristic for solid electrodes. (Hg(Ag)FE) is widely use both for voltammetric analysis of organic and inorganic compounds content in different samples with low detection limits.[53–60] The electrode preparation allows its functioning in a stable way for numerous regeneration cycles for several months, which was proved in a previously published paper.[61] The detailed mechanical construction of the Hg(Ag)FE used in this experiments is described in Ref. 62.

In this study new, high sensitive voltammetric method of betamethasone determination is presented. Renewable silver based amalgam film electrode has been used as the working electrode due to its properties that allow to obtain high precision, repeatability and high sensitivity of the method. New procedure was successfully applied for betamethasone detection in popular pharmaceutical samples.

Experimental

Measuring apparatus.—Voltammetric measurements were performed on the multipurpose Electrochemical Analyzer M161 with the electrode stand M164 (both MTM-ANKO, Poland) and the EA-GRAPH software. The classical three-electrode quartz cell (volume of 20 mL) were used with the silver based amalgam film electrode (Hg(Ag)FE) as the working electrode, a silver reference electrode Ag/AgCl/KCl (3 mol L^{-1}) and a platinum rod as an auxiliary electrode. Stirring was performed using a magnetic Teflon coated bar, rotating at approximately 500 rpm. pH measurements were performed with laboratory pH-meter (N-512 elpo, Polymetron, Poland). All experiments were carried out at room temperature.

Chemicals.—Standard stock solution (10 mL) of 0.01 mol L^{-1} betamethasone (Sigma Aldrich) was obtained by dissolution of an appropriate amount of this reagent in equal proportion of water and ethanol (POCH) and then stored in fridge. Acetate buffer (0.5 mol L^{-1}, pH 3.0) was prepared by mixing the appropriate amounts of glacial acetic acid (Suprapur, Merck) and crystalline sodium acetate anhydrous (Sigma Aldrich). Water free anionic surfactant Marlinat 242/90 MC (MIPA-Laureth Sulfate and Propylene Glycol) was obtained from Sasol. Triton X-100 was purchased from Windsor Laboratories Ltd.

All reagents were of analytical grade and used without further purification. All solutions were prepared with double-distilled water.

Sample preparation.—Pharmaceutical formulations - suspension for injections (Diprophos, MSD Polska) and transdermal sprays (Belosalic, Belupo, and Salbetan, Sun-Farm) containing 7 mg/ml and 0.5 mg/g of betamethasone respectively, were obtained from local pharmacy. For the analysis of betamethasone content in proposed samples, the proper amount of the was transferred to the 5 mL volumetric flask and filled with ethanol and water in a 1:1 ratio in order to get required concentration. After complete homogenization by ultrasounds, solutions were ready to analyze.

The content of betamethasone in pharmaceutical samples was determined by the standard addition method and its suitability was validated using recovery parameter.

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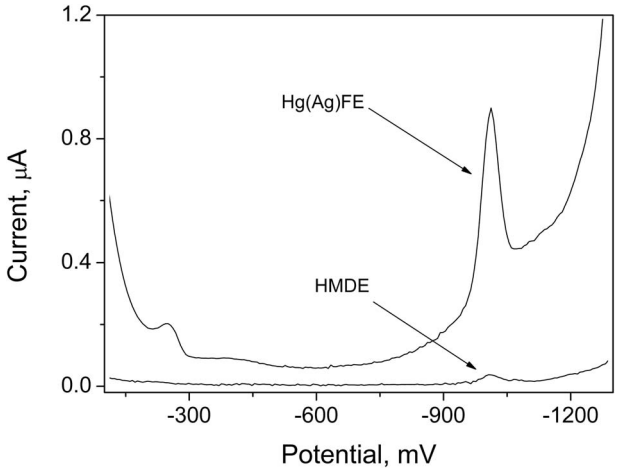


Figure 1. Comparison of DP voltammograms obtained for 0.2 μM betamethasone in 0.1 M acetate buffer (pH 3.0) for HMDE and Hg(Ag)FE electrode. The electrode areas were 10.3 mm^2 for the (Hg(Ag)FE) and 1.6 mm^2 for the HMDE.

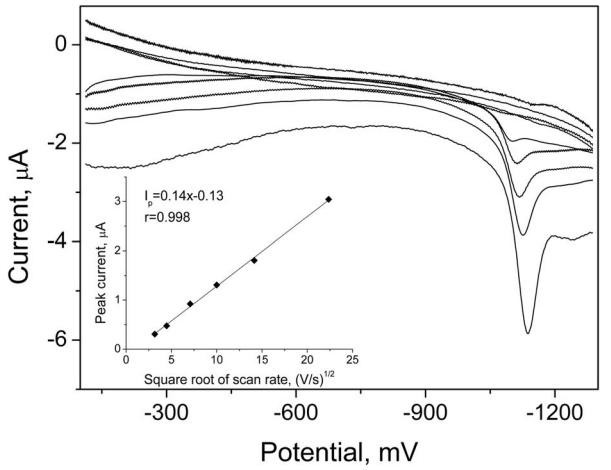


Figure 2. Cyclic voltammograms for $1 \cdot 10^{-5}$ mol L^{-1} betamethasone in 0.1 mol L^{-1} acetate buffer (pH 3.0) measured for the scan rate of: 0.02; 0.05, 0.1, 0.2 and 0.5 V s^{-1} respectively, measured on an amalgam film electrode Hg(Ag)FE.

Standard procedure of measurements.—Quantitative voltammetric measurements of betamethasone were performed using differential pulse adsorptive stripping voltammetry (DPV). The mechanism of BM reduction reaction was investigated by linear sweep voltammetry (LSV). All measurements were performed on the silver based amalgam film electrode (Hg(Ag)FE) with refreshable surface. As the supporting electrolyte, the solution consisted of 0.1 mol L^{-1} acetate buffer with pH of 3.0 and 5 μL of 0.1% Marlinat was used (total volume 10 mL). The procedure of measurement was performed following sequence of three steps: (a) the conditioning of (Hg(Ag)FE) electrode by applying potential of −1300 mV for 5 s (b) the betamethasone preconcentration step: $E_{acc} = -100$ mV; $t_{acc} = 10$ s; (c) recording voltammograms in the potential range from −100 mV to −1300 mV after the rest period of 3 s.

Other conditions for the DP mode were as follows: step potential 6 mV; pulse amplitude 50 mV; time step potential 20 ms (10 ms waiting and 10 ms sampling time).

Results and Discussion

Voltammetric behavior of betamethasone on Hg(Ag)FE electrode.—Voltammetric DPV technique and an amalgam film electrode (Hg(Ag)FE) have been chosen for low betamethasone concentrations measurements. The Hg(Ag)FE electrode were proposed due to its higher sensitivity and smaller use of mercury during measurements. To prove this, the BM signals were registered both on HMDE and Hg(Ag)FE and presented in Fig. 1.

Betamethasone signals obtained on (Hg(Ag)FE) was 3.6 times higher than for HMDE, considering the size of electrodes working surface. Thus, the choice of an amalgam film electrode for further studies was confirmed due to better properties of BM peak. The BM peak potentials values were equal to −1012 mV and −1006 mV for Hg(Ag)FE and HMDE respectively.

Betamethasone signal measured on the Hg(Ag)FE surface grew linearly with the increasing size of working surface from 4.9 mm^2 to 13.9 mm^2, according to the following equation:

$$I_p = 0.048\,(\pm 0.0009)\left[\mu A mm^{-2}\right] + 0.11\,(\pm 0.009)\,[\mu A]\,,$$
$$r = 0.998 \qquad [1]$$

For further studies the electrode surface of 10.3 mm^2 was chosen in order to obtain well shaped and stable betamethasone signals and good ratio of signal to background current.

Cyclic voltammetry.—Betamethasone reduction process was investigated using linear sweep voltammetry. The voltammograms of 10 μmol L^{-1} BM reduction peak measured for the scan rate in the range from 20 to 500 mV s^{-1} are presented in Figure 2. Due to the absence of betamethasone oxidation peak in the anodic scan it is possible to said, that BM reduction process on the Hg(Ag)FE is irreversible. Obtained CV scans were also used during the process of BM reduction mechanism investigation. The dependencies between BM peak current versus scan rate and square root of the scan rate were crossed out. Linear response was obtained between the BM peak current and the square root of the scan rate, and described as in Equation 2:

$$I_p = 0.14x - 0.13\ \mu A,\ r = 0.998 \qquad [2]$$

Considering obtained results, it can be said that betamethasone reduction reaction is a diffusion controlled process.

In order to propose the mechanism of betamethasone reduction, DPV scans with different supporting electrolyte pH values and CV scans with different scan rate were performed. Increase of the scan rate value resulted in the change of the BM cathodic peak potentials, which were shifting toward the more negative values. Dependence of the peak potential vs. natural logarithm of scan rate results in a straight line with linear regression equation of:

$$E_k = -0.013 \ln v - 1.06\,[V]\,,\ r = 0.997 \qquad [3]$$

Considering the theory of an irreversible electrode reaction, from the slope of Ek vs. ln(v), the value of αn could be calculate as 1.01. Therefore assuming α as 0.5, the number of electrons transferred in BM reduction process could be calculate as 2.

In order to clarify the betamethasone reduction mechanism, pH influence on BM peak was investigated. In the Fig. 3 the dependence between the betamethasone peak potential and current on the acetate buffer pH with values from 3.0 to 5.0 is presented. The changing supporting electrolyte pH strongly influenced both BM potential and peak current values, which manifested the best stability and repeatability in the acetate buffer with pH value of 3.0. Therefore, acetate buffer of pH 3.0 was used in further measurements.

The less acidic pH of supporting electrolyte, the more betamethasone peak potential was shifting toward the more negative values. Obtained dependence between BM peak potential vs pH value is linear and can be described as in Equation 4.

$$E_p = -0.057\ pH - 0.83\ [V] \qquad [4]$$

The slope of 0.057 V/pH assume that the number of electrons exchanged during the BM reduction process is equal to the number of exchanged protons. Mechanism of possible betamethasone reduction on an amalgam film electrode is presented in Scheme 1.

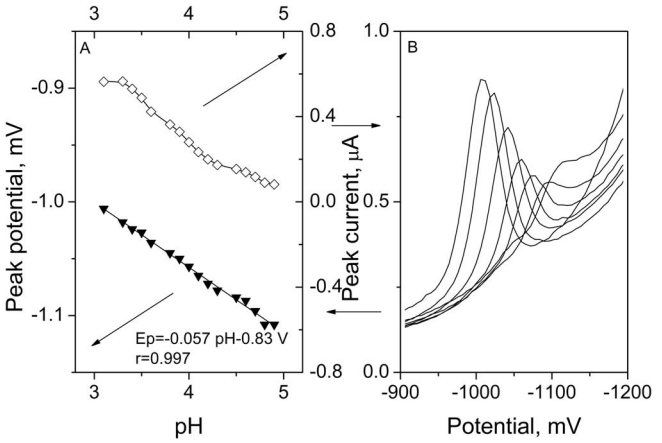


Figure 3. Comparison of dependence of peak potential and current on pH (A) and voltammograms (B) obtained for 0.2 μM betamethasone, measured in the range from 3.0 to 5.0 in 0.1 mol L^{-1} acetate buffer.

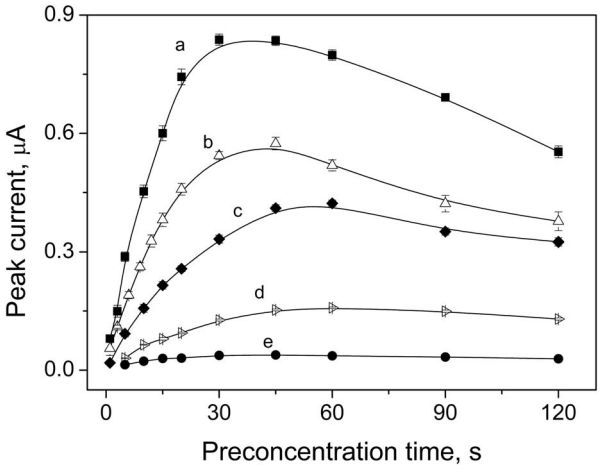


Figure 4. Dependence of the peak current on betamethasone preconcentration time in the range from 0 to 120 s for: (a) 0.2; (b) 0.1; (c) 0.05; (d) 0.01 and (e) 0.005 $\cdot 10^{-6}$ mol L^{-1} betamethasone concentration in 0.1 mol L^{-1} acetate buffer (pH 3.0).

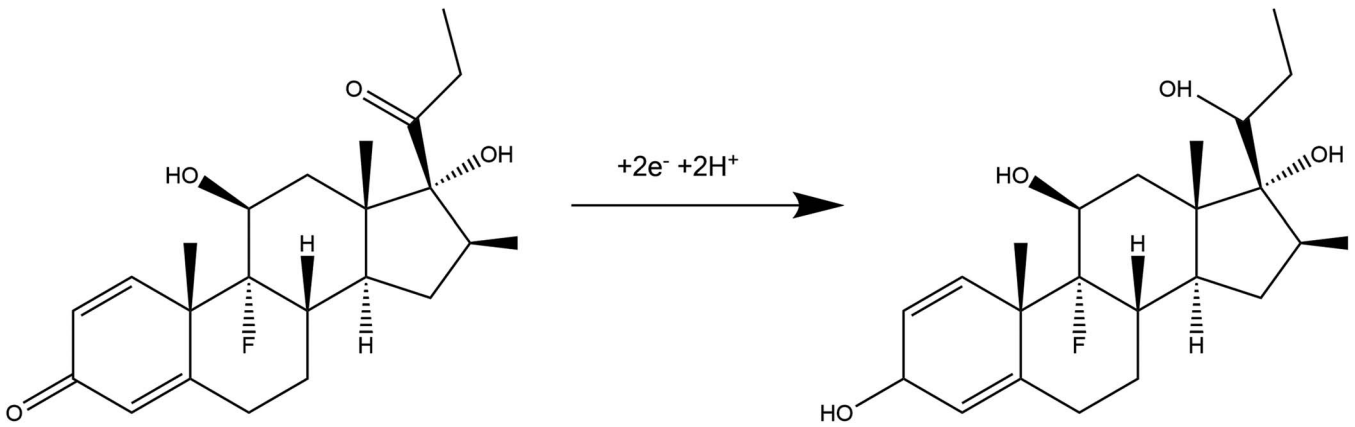
Influence of DPV parameters on betamethasone peak.—In order to establish the best conditions for high sensitive betamethasone determination, the DPV instrumental parameters were investigated. To choose optimal values for betamethasone determination, parameters such as pulse amplitude (ΔE), step potential (E_s), sampling (t_s) and waiting (t_w) time were tested in the wide range of values. For further measurements following values were chosen: pulse amplitude of 50 mV, step potential of 6 mV, waiting and sampling time of 10 ms each.

Influence of preconcentration time and potential on betamethasone peak.—In order to provide the best possible instrumental parameters for high sensitive betamethasone determination, preconcentration time and potential were investigated in the wide range of valued due to its high impact on sensitivity of voltammetric methods. The preconcentration potential was studied in the range from 100 to −650 mV (supplementary file 1). In the range from −50 to −650 mV, the betamethasone peak current values were stable, with the average value of about 0.55 μA. When the preconcentration potential was set as 0 mV or in the positive values (50–100 mV), the betamethasone peak was strongly influenced by the change, decreasing in its size. Thus for further measurements the preconcentration potential of −100 mV has been chosen due to best properties of obtained BM peak.

Influence of preconcentration time on the betamethasone peak was studied in the range from 0 s to 120 s, for different BM concentration (Fig. 4). For higher BM concentrations (0.2, 0.1 $\cdot 10^{-6}$ mol L^{-1}) the peak current value decrease rapidly after reaching its maximum value. For lower BM concentrations (0.05, 0.01, 0.005 $\cdot 10^{-6}$ mol L^{-1}) the change in peak current value is more gentle after reaching its maximum. Therefore, the optimal preconcentration time for betamethasone is strongly connected with its concentration, and its choice should be always based on its assumed concentration in examined sample.

Influence of base electrolyte composition on betamethasone peak.—In order to select optimal environment for sensitive betamethasone determination, its electrochemical behavior has been investigated in different electrolytes such as: 0.1 mol L^{-1} dihydrogen phosphate (V) of potassium (pH 4.6), 0.1 mol L^{-1} dipotassium phosphate (pH 9.1), 0.1 mol L^{-1} sodium tetraborate solution (pH 10.0), 0.1 mol L^{-1} ammonia buffer (pH 8.2), 0.1 mol L^{-1} phosphate buffer (pH 6.7 and 5.5) and 0.1 mol L^{-1} acetate buffer (pH 5.6 and 3.0) (Fig. not included). The best BM reduction signals considering the value of background current, repeatability of the peak and sensitivity of the measurements were obtained in acetate buffer with the pH value of 3.0. Therefore for the further analysis acetate buffer with pH of 3.0 was chosen.

Besides the type of supporting electrolyte, the influence of its concentration on the betamethasone peak was investigated. Concentration of the acetate buffer with pH 3.0 was tested in the range from 0.0125 to 0.1125 mol L^{-1}. Change in the concentration of supporting electrolyte did not affect the value BM peak current, but with its higher concentration the background current value was decreasing, which was favorable phenomenon for further analysis. Therefore, the



Scheme 1. Possible scheme of betamethasone reduction on an amalgam film electrode.

Table I. Betamethasone peak current change in presence of interferents.

Interferent\concentration of interferent, μM	Peak current value, μA							
	0	2	0.2	20	166	10000	$1.7 \cdot 10^5$	Signal change,%
Mn	0.55	-	-	-	-	-	-	-
Cu	0.56	-	-	-	-	-	-	-
Pb	0.56	-	-	-	-	-	-	-
Co	0.53	0.47	-	-	-	-	-	10
Zn	0.50	-	-	-	-	-	-	-
Fe	0.55	-	-	-	-	-	-	-
Ni	0.53	-	-	-	-	-	-	-
Mo	0.52	-	-	-	-	-	-	-
Ascorbic acid	0.55	-	-	-	-	-	-	-
Glucose	0.56	-	-	-	-	-	-	-
Acetylsalicylic acid	0.54	-	-	-	-	-	-	-
Citric acid	0.53	-	-	0.29	-	-	-	45
Adrenalin	0.55	-	-	-	-	-	-	-
Testosterone	0.54	0.32	-	-	-	-	-	40
Spironolactone	0.54	0	-	-	-	-	-	100
Dexamethasone	0.56	0	-	-	-	-	-	100
Triton X-100	0.56	-	-	-	-	-	0.15	71

acetate buffer concentration value of 0.1 mol L^{-1} has been chosen for measurements.

Interferences.—The examined ions, such as Cu (II), Pb (II), Ni (III), Fe (III), Zn (II), Mo (VI) and Mn (II) ($2 \cdot 10^{-6}$ mol L^{-1} added) did not interfere with betamethasone peak. Only Co (II) ions ($2 \cdot 10^{-6}$ mol L^{-1} added) caused decrease in the BM peak current value of about 10%. Also organic compounds such as ascorbic acid ($2 \cdot 10^{-6}$ mol L^{-1} added), glucose ($50 \cdot 10^{-3}$ g mL^{-1}), acetylsalicylic acid (0.1 mol L^{-1} added) and adrenalin ($2 \cdot 10^{-7}$ mol L^{-1} added) were tested and did not affect the betamethasone peak. However, it turned out that the citric acid strongly interfere with betamethasone peak, and addition of $2 \cdot 10^{-5}$ mol L^{-1} caused BM peak decrease of about 45%. Also addition of testosterone ($2 \cdot 10^{-6}$ mol L^{-1}) cause BM peak decrease of 40%. Addition of dexamethasone and spironolactone ($2 \cdot 10^{-6}$ mol L^{-1} each) makes quantitative analysis of betamethasone unable due to superposition of the peaks (in the presence of dexamethasone) or complete disappearance of BM signal (in the presence of spironolactone). Surface active compound, Triton X-100, was tested due to its high impact on reduction signal at mercury electrodes. Triton X-100 was tested in a range of concentration from $0.05 \cdot 10^{-6}$ g mL^{-1} to $0.3 \cdot 10^{-6}$ g mL^{-1} and for its maximum concentration betamethasone peak was decreasing in size of about 71%. In Table I changes in betamethasone peak current before and after interferents dosing are presented.

Analytical performance.—The DPV betamethasone calibration curves on amalgam film electrode are presented in Fig. 5.

Betamethasone signal was linear in the wide range of concentration values, between $5 \cdot 10^{-9}$ and $0.8 \cdot 10^{-6}$ mol L^{-1}. The lowest BM detection limit was obtained for preconcentration time of 45 s and it was equal to $1.6 \cdot 10^{-9}$ mol L^{-1}, with linearity up to $0.1 \cdot 10^{-6}$ mol L^{-1}. The reproducibility of the method measured and specified as RSD was 2.1% (with n = 5, $0.1 \cdot 10^{-7}$ mol L^{-1} BM concentration). Comparison of betamethasone detection limits for its different determination methods is enclosed in the Table II.

To validate the method drugs samples were investigated in this respect.

Betamethasone determination was performed in injections and transdermal sprays available commercially in pharmacies. Sample preparation and measurements were prepared and performed according to the procedure from point 2.4.

Results of the betamethasone determination in different medications are presented in Tables III (with recovery). Measured recovery was very good, in the range of 95 and 105%. Therefore, the analytical usefulness of BM determination in injections and transdermal sprays measured on an amalgam film electrode with stripping voltammetry techniques was confirmed.

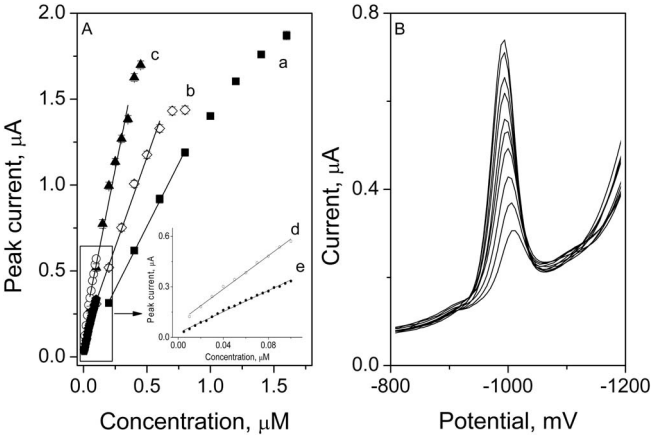


Figure 5. DPV betamethasone calibration curves obtained for preconcentration times: (a) 5 s; (b) 10 s; (c) 20 s; (d) 30 s and (e) 45 s in 0.1 mol L^{-1} acetate buffer (pH 3.0) (A) and corresponding voltammograms for preconcentration time 30 s and concentration in the range from 0.01 to 0.1 mol L^{-1} (B).

Conclusions

In this work new method of betamethasone determination on silver based amalgam film electrode has been proposed. Due to high sensitivity of DPV technique and a beneficial features of the electrode it was possible to measure betamethasone in nanomolar concentration with the detection limit of $1{,}6 \cdot 10^{-9}$ mol L^{-1} ($6.28 \cdot 10^{-6}$ g L^{-1}) for the preconcentration time of 45 s. The reproducibility measured as RSD is very good, for concentration of $0.1 \cdot 10^{-7}$ mol L^{-1} is equal to 2.1% (n = 5). The new procedure was successfully applied for betamethasone determination in different types of drug samples with very good recovery, calculated in the range from 95% to 105%. The proposed method exhibits high sensitivity and accuracy of performed measurements with short time of analysis, that indicates its suitability for routine quality control analysis.

Table II. Comparison of betamethasone determination method.

Method	Linear range	Detection limit	Source
HPLC[1]	0.04 – 0.2 mg mL^{-1}	$7.00 \cdot 10^{-9}$ g L^{-1}	36
MEKC[2]	0.03 – 0.2 mg mL^{-1}	$0.34 \cdot 10^{-6}$ g L^{-1}	36
HPLC–ESI-MS-MS[3]	3 – 500 ng mL^{-1}	$0.013 \cdot 10^{-6}$ g L^{-1}	37
LC–LC–MS–MS[4]	1 – 50 ng mL^{-1}	$0.20 \cdot 10^{-6}$ g L^{-1}	39
RP HPLC[5]	2 – 10 μg mL^{-1}	$0.12 \cdot 10^{-9}$ g L^{-1}	40
HPLC–ESI–MS–MS[3]	0.1 – 50 ng mL^{-1}	$0.1 \cdot 10^{-7}$ g L^{-1}	46
DPP[6]	1.5 – 43 mg mL^{-1}	$1.06 \cdot 10^{-3}$ g L^{-1}	47
SWV[7]	0.19 – 39.2 μg mL^{-1}	$9.81 \cdot 10^{-8}$ g L^{-1}	48
OSWV[8]	0.39 – 9.81 μg mL^{-1}	$1.96 \cdot 10^{-7}$ g L^{-1}	49
Spectrophotometry	0 – 20 μg mL^{-1}	$0.14 \cdot 10^{-3}$ g L^{-1}	63
DPV[9]	1.96 – 314 μg mL^{-1}	$6.28 \cdot 10^{-7}$ g L^{-1}	This work

[1]High Performance Liquid Chromatography.
[2]Micellar Electrokinetic Chromatography.
[3]High Performance Liquid Chromatography Electrospray Ionization Tandem Mass Spectrometry.
[4]Coupled Column Liquid Chromatographic Tandem Mass Spectrometry.
[5]Reversed Phase High Performance Liquid Chromatography.
[6]Differential Pulse Polarography.
[7]Square Wave Voltammetry.
[8]Osteryoung Square Wave Voltammetry.
[9]Differential Pulse Voltammtery.

Table III. Results of betamethasone determination in different pharmaceutical samples.

	Betamethasone found ± (recovery,%),		
Betamethasone added	Diprophos[1] mg/mL	Belosalic[2] mg/g	Salbetan[2] mg/g
0	9.48 ± 0.9	0.51 ± 0.04	0.54 ± 0.08
1 mg	-	1.06 ± 0.02 (97)	1.07 ± 0.03 (105)
1.5 mg	-	1.47 ± 0.01 (95)	1.49 ± 0.05 (99)
18 mg	17.4 ± 0.8 (100)	-	-
36 mg	34.7 ± 1.3 (101)	-	-
52 mg	52.3 ± 0.9 (100)	-	-

[1]9 mg/ml of betamethasone declared.
[2]0.5 mg/g of betamethasone declared.

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